

AI-Assisted Audit Data Quality & Reconciliation Platform

Full Design Document (Competition Version)

1. Project Overview

This project proposes a web-based AI-assisted audit platform designed to streamline data validation, reconciliation, and audit documentation. Using Python and IBM Bob, the platform enables auditors to automate repetitive data tasks while maintaining full control and transparency.

2. Problem Statement

Audit teams face inefficiencies due to manual data validation, inconsistent data quality checks, and time-consuming documentation processes. Common issues include missing values, duplicate records, schema mismatches, and unclear transformation logic.

3. Solution Overview

The platform provides an end-to-end workflow from data upload to audit report generation. It integrates Python-based data processing with AI-assisted explanation and documentation.

Workflow:

Upload → Profile → Validate → Reconcile → Analyze → Generate Report

4. Core Features

- File Upload (CSV/Excel)
- Automated Data Profiling
- Data Quality Rule Engine
- Source-to-Target Reconciliation
- AI-Powered Explanation Layer
- Audit Workpaper Export

5. Data Profiling Capabilities

- Row/Column counts
- Missing values detection
- Duplicate detection
- Data type validation
- Unique value counts
- Outlier identification

6. Data Quality Rule Engine

Rules:

- Required fields not null
- Unique primary keys
- Valid date formats
- Value range validation
- Referential integrity checks

7. Reconciliation Engine

Checks:

- Record count comparison
- Missing/extra records
- Column mapping mismatches
- Data type inconsistencies

8. AI Agent Functions

- Explain anomalies in plain English
- Suggest column mappings
- Generate audit findings
- Draft audit documentation
- Assist developers via IBM Bob

9. System Architecture

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Frontend (Streamlit)
↓
Python Backend
↓
Data Processing (pandas)
↓
AI Layer (IBM Bob)
↓
Report Generation
```

10. Technology Stack

Frontend: Streamlit
Backend: Python
Libraries: pandas, numpy
AI: IBM Bob
Output: Excel / HTML

11. User Workflow

1. Upload datasets
2. Run profiling
3. Execute data quality checks
4. Perform reconciliation
5. Review AI-generated insights
6. Export audit report

12. Example Scenario

Source: 10,000 rows
Target: 9,850 rows
Findings:

- 150 missing records
- 138 null values
- Mapping mismatch detected

13. Team Structure

Member 1:

- Backend development
- Data processing logic

Member 2:

- UI/UX design
- Presentation and documentation

14. IBM Bob Integration

IBM Bob is used as an AI development assistant to accelerate coding, testing, debugging, and documentation. It enables a small team to deliver a scalable solution efficiently.

15. Judging Criteria Alignment

Completeness: Full workflow implemented. Feasibility: Built using proven technologies. Creativity: AI applied to audit workflow. Usability: Simple UI. Effectiveness: Reduces manual work and improves accuracy.

16. Scalability & Future Work

- Database integration
- Role-based access
- Real-time monitoring
- Advanced anomaly detection
- Enterprise audit workflow integration

17. Success Metrics

- Time saved in audit tasks
- Number of issues detected
- Reduction in manual effort
- User adoption rate

18. Conclusion

This platform demonstrates how AI and Python can transform audit workflows by improving efficiency, accuracy, and scalability. It provides a strong foundation for future enterprise-level audit systems.